SXM GO

Gamified Island Exploration

Taylor Knipe

Kaiden Vandegrift

Oliver Skoczylas

Paul Han

SPRINT 2

PRESENTATION

TEAM MEMBERS

- Oliver Skoczylas (Dev 1) – Core Auth & User Management
- Kaiden Vandegrift (Dev 2) – Map, Location & Check-In
- Taylor Knipe (Dev 3) – Gamification Engine
- Paul Han (Dev 4) – Social Features & Leaderboards

https://sxmgo.vercel.app/
https://github.com/OliverSkoczylas/SXM-GO

PROJECT FOCUS

THE PROBLEM

St. Maarten has 37 beaches, 2 nations, and countless local businesses — yet tourists explore only a fraction of what the island offers.

Limited awareness and no structured discovery mechanism leaves local businesses underserved and visitors missing out.

There is no engaging, interactive way for tourists to discover hidden gems.

THE SOLUTION

- GPS-based check-ins to verify presence at 50+ locations
- Points, badges & leaderboards that reward exploration
- Compete globally and within travel groups
- Curated itineraries: Foodie, Beach Hopper, Cultural
- Drives foot traffic to local businesses

TARGET: Cruise passengers · Resort guests · Independent travelers · Local businesses seeking foot traffic

PRIORITIZED SPRINT 2 GOALS

P1

Social OAuth (Google & Apple)

Complete OAuth 2.0 flow for Google and Apple; token storage in Keychain/Keystore; integrate Supabase Auth provider

Owner: Dev 1

P1

Offline Map Caching & QR Fallback

Cache 50+ location tiles for offline access; QR code check-in verification at poor-GPS zones; 50+ pins fully seeded with category icons

Owner: Dev 2

P1

Badge Unlock & Challenge Integration

Connect check-in events to badge unlock logic; live challenge progress frontend for all 4 badges (Foodie, High Roller, Island Nomad, Sun Chaser)

Owner: Dev 3

P2

Weekly & Monthly Leaderboards

Weekly leaderboard resetting every Monday; monthly leaderboard; push notification infrastructure (FCM/APNs setup)

Owner: Dev 4

P2

Advanced GPS Anti-Fraud

Owner: Dev 2

P3

TLS Rate Limiting & Profile Update UI

Owner: Dev 1

SPRINT 1 REVIEW · SPRINT 2 KICKOFF

✓ Auth System Live

COMPLETE

Email/password registration, login, JWT session management, and user profile CRUD all operational in Supabase.

Dev 1

✓ Map Rendering (50+ Pins)

IN PROGRESS

OSM integration live with 10 pins from Sprint 1; expanding to full 50+ seeded locations with category icons in Sprint 2.

Dev 2

✓ Check-In Logic

COMPLETE

GPS-based 50m radius check-in verified; duplicate prevention within 24h; points awarded within 500ms of check-in.

Dev 2

✓ Points Engine Backend

COMPLETE

Base point tracking, duplicate request handling, and challenge progress tracking operational on DB side.

Dev 3

✓ Global Leaderboard

COMPLETE

Top-100 real-time leaderboard with Supabase Realtime; updates within 2 seconds of any point event.

✓ Sponsor Feedback

NOTED

Sponsors responded positively to MVP scope. Sprint 2 demo to showcase full check-in → badge flow end-to-end.

PROJECT BACKLOG · SPRINT BACKLOG

PROJECT BACKLOG (Selected Items)

FR-001–011	User Auth & Profiles	DONE
FR-012–026	Interactive Map & Locations	DONE
FR-027–035	🔒 GPS Check-In + Anti-Fraud	IN PROGRESS
FR-036–043	Points & Scoring System	IN PROGRESS
FR-059–071	Badges & Challenges	IN PROGRESS
FR-072–075	Leaderboard System	IN PROGRESS
TR-022–026	🔒 Security (TLS, Rate Limit)	IN PROGRESS
FR-044–058	Itinerary System	BACKLOG
FR-084–096	Activity / Route Tracking	BACKLOG
FR-097–103	Social Features & Friends	BACKLOG

SPRINT 2 BACKLOG

○ 🔒 Social OAuth (Google & Apple)	Dev 1
○ 🔒 TLS/HTTPS rate limiting — all endpoints	Dev 1
○ Profile update UI & password change	Dev 1
○ 50+ location pins seeded & categorised	Dev 2
○ Offline map tile caching	Dev 2
○ 🔒 QR code fallback check-in	Dev 2
○ 🔒 Advanced GPS spoofing detection	Dev 2
○ Badge unlock & in-app notification	Dev 3
○ Challenge tracking frontend (4 badges)	Dev 3
○ Weekly leaderboard (Monday reset)	Dev 4
○ Monthly leaderboard	Dev 4

🔒 = *Cybersecurity-related item*

CYBERSECURITY ASSESSMENT

Authentication Security — IMPLEMENTED

- Passwords hashed via bcrypt (industry standard)
- JWTs stored in Keychain (iOS) / Keystore (Android)
- Sessions expire after 30 days of inactivity
- Rate limiting on login endpoint (5 attempts/min)

Data Privacy — IN PROGRESS

- HTTPS/TLS 1.3 now active on all endpoints (Sprint 2)
- Location data collected only with explicit permission
- GDPR/CCPA compliance review scheduled Sprint 3
- User data deletion endpoint operational

GPS Anti-Fraud — IN PROGRESS

- Basic spoofing detection via mock location flags
- Impossible travel speed pattern detection added
- QR code fallback now live for poor GPS zones
- Manual review queue for flagged check-ins

API Security — IN PROGRESS

- API keys obfuscated in mobile bundle
- Input validation and sanitisation on all endpoints
- SQL injection protection via Supabase ORM
- Rate limiting expanded to all routes (Sprint 2)

NOT COMPLETED · LESSONS LEARNED

NOT COMPLETED THIS SPRINT

Itinerary Builder UI (Dev 4)

Weekly leaderboard took priority; data model is done, full UI deferred to Sprint 3 | Dev 4

Push Notification Delivery (Dev 4)

FCM/APNs infrastructure begun; full delivery pipeline pushed to Sprint 3 | Dev 4

Activity Route Tracking (Dev 3)

GPS route recording deprioritised; core badge unlock flow took precedence | Dev 3

Profile Update UI (Dev 1)

OAuth integration consumed Sprint 2 capacity; deferred to Sprint 3 | Dev 1

Facebook OAuth (Dev 1)

Post-MVP feature; Google/Apple login prioritised per requirements document | Dev 1

LESSONS LEARNED

1. OAuth Config Takes Longer Than Estimated

Provider edge cases (Apple on Android) require extra time. Budget 1.5× the estimate for OAuth work.

2. Seed Data Quality Matters Early

Placeholder pins slowed Dev 2. Real coordinates should be seeded before frontend pin work starts.

3. Security Must Be Sprint 1 Scope

TLS rate limiting on all routes should have been in Sprint 1. Retrofitting security costs more than building it in.

4. Integration Tests Before Demo

Check-in → points → badge pipeline had edge cases found late. Integration tests before sprint review are mandatory.

5. Supabase Realtime Tier Planning

Free tier connection limits hit during load test. Upgrading to paid tier before Sprint 3 beta is required.

TEAM CONTRIBUTIONS

Developer 1 – Oliver Skoczylas

Core Auth & User Management

Completed:

- Social OAuth (Google & Apple) — live
- TLS rate limiting on all auth routes
- GDPR data controls — partial
- Account deletion endpoint

Next: Profile update UI, Facebook OAuth (post-MVP)

Developer 2 – Kaiden Vandegrift

Map, Location & Check-In

Completed:

- 50+ location pins seeded with categories
- Offline map tile caching live
- QR code fallback check-in live
- Advanced GPS spoofing pattern detection

Next: Location detail view polish, photo verification (optional)

Developer 3 – Taylor Knipe

Gamification Engine

Completed:

- Badge unlock logic connected to check-in events
- In-app notification on badge earn
- Challenge progress frontend (all 4 badges)
- Bronze/Silver/Gold tier models

Next: Activity route tracking, streak bonus frontend

Developer 4 – Paul Han

Social Features & Leaderboards

Completed:

- Weekly leaderboard (Monday reset) live
- Monthly leaderboard operational
- Push notification infrastructure (FCM setup)
- Itinerary data model complete

Next: Itinerary builder UI, group competition, friend system

LIVE DEMO

Sprint 2 Feature Walkthrough

01

Social OAuth
(Google Login)

02

Offline Map &
QR Check-In

03

Badge Unlock
Notification

04

Weekly
Leaderboard